

Aman Nindra

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Education

University of California, Merced

Expected May 2027

Bachelors of Science in Computer Science & Engineering

• **Relevant Coursework:** Data Structures, Advanced Programming, Linear Algebra, Calculus I-III, Probability & Statistics

• **Clubs:** Machine Learning Club & Data Science Society

Skills

- **Languages:** Python, JavaScript, TypeScript, C++, HTML/CSS, TailwindCSS, SQL
- **AI/ML:** PyTorch, Scikit-Learn, OpenCV, NumPy, Pandas, YOLO, Transformers, Natural Language Processing, Computer Vision
- **Cloud & Tools:** AWS (SageMaker, S3), Firebase, Flask, FastAPI, Next.js, React.js, Docker, Git, Linux, VS Code, Cursor
- **Courses:** IBM Computer Vision, Stanford Supervised Machine Learning, JSE Certified Entry-Level JavaScript, Introduction to Selenium, Facial Expression Recognition with PyTorch

Experience

Software Engineering Intern

Sep 2025 – Dec 2025

Arrowz (Mental Health Startup)

Remote

- Shipped cross-platform Flutter/Dart app for emotion tracking across **25+ screens**; modular UI components reduced code duplication by **40%**.
- Standardized state management and responsive layouts; improved build stability and developer velocity across the team.

Mentor, AI Code Agent

Nov 2025 – Present

Machine Learning Club

University

- Scoped and led **Code Companion**, an AI-powered Python debugging tool that ingests GitHub repos and suggests LLM-generated fixes with explanations.

Mentor, Research Copilot

Nov 2025 – Present

Data Science Society Club

University

- Managed a RAG-based research assistant that ingests PDFs and arXiv links and answers questions with citation-grounded quotes.
- Designed system architecture (**Next.js**, **FastAPI**) and evaluation plan focusing on answer quality, latency, and pilot user feedback.

Prompt Engineer

Sep 2023 – Dec 2024

Hotpot.ai (AI Startup)

Remote

- Designed and refined prompts with guardrails for Hotpot.ai's custom image generation models; produced high-quality visuals published on social platforms X/Twitter and Reddit.
- Collaborated with Machine Learning engineers to integrate emerging image generation capabilities and socialize for global users.
- Managed the stock market account on Twitter, publishing release updates and market insights to grow audience engagement.

Projects

Self-Hosted Distributed AI Training Platform — PyTorch, Kubernetes, FastAPI, React.js, Linux

Feb 2026 – Present

- Designing and building a self-hosted AI training platform to enable users worldwide to upload datasets (S3 Style), configure hyperparameters, and train models through a web interface.
- Implementing GPU-aware job scheduling to dynamically allocate training workloads across multiple Linux machines based on real-time VRAM and compute availability using Kubernetes.
- Enabling concurrent model training on a single GPU when memory constraints permit, maximizing hardware utilization through resource isolation and process-level management.
- Deploying a Raspberry Pi as a lightweight control node to host the website, manage job queues, and persist uploaded datasets to an attached external storage device.

Autonomous Bicycle — ROS 2, Python, Jetson Nano

Feb 2026 – Present

- Developing an autonomous electric bicycle capable of navigating real-world terrain without human control.
- Integrating ROS 2 on a Jetson Nano Super (edge AI GPU) for perception, localization, and control.

AI Emotion Intelligence Engine — PyTorch, Transformers, Scikit-learn, AWS SageMaker — [GitHub](#)

Jan 2026

- Fine-tuned **BERT-base** for **28-class** multi-label emotion detection; achieved **Macro F1: 0.64–0.70** and **Jaccard (samples): 0.59–0.67**.
- Built **per-emotion threshold calibration** (0.01–0.99 grid search per label), replacing a single global cutoff and improving minority-emotion performance.
- Benchmarked against GoEmotions baselines (Macro F1 ~0.49) and achieved **Macro F1 ~0.64**, **Precision ~0.52**, and **Recall ~0.84** in 15 epochs.

Wildlife Camera Trap Classification — PyTorch, Vision Transformers, AWS SageMaker — [GitHub](#)

Jan 2026

- Achieved **Top 4** ranking (log loss: **0.6695**) using **EVA-Large** (300M parameters) Vision Transformer on 16.5K camera trap images across 8 species.
- Engineered production-grade training pipeline with **DistributedDataParallel (DDP)** across **4 Tesla V100 GPUs** on AWS SageMaker, implementing mixed-precision training, gradient clipping, and class-weighted loss.
- Optimized model generalization achieving **-0.64% train-val gap** (91.27% val accuracy) through advanced Albumentations augmentation and ImageNet-22K pre-training; outperformed baseline ResNet18 by **3.1%**.

AI-Resume Analyzer — Python, Flask, React.js, TypeScript, AWS, OpenAI API — [GitHub](#)

June - July 2025

- Containerized full-stack app that ingests PDFs and returns resume recommendations, jobs that match your resume, and a personalized dashboard to manage saved job listings.
- Engineered a Firebase Realtime Database caching layer for job queries, reducing redundant third-party web scraper (Apify) API calls and significantly improving frontend response times.
- Designed a daily background cron scheduler (APScheduler) to auto-refresh job listings, implementing MD5 signature hashing to detect and prevent duplicate database entries.